

DISCRETE FOURIER TRANSFORM

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ABSTRACT. Part of the course notes for this course. Sergey's notes for his lectures on the discrete Fourier transform.

1. THE GROUP OF CHARACTERS FOR A

Consider a finite Abelian group A . Recall that a group map $\chi : A \rightarrow \mathbb{C}^*$ is called a character of A . Notice that since for every $a \in A$ there exists $s : a^s = 1$, the absolute value $|\chi(a)| = 1$, i.e. the image of χ belongs to the circle subgroup $S = \{\alpha \in \mathbb{C} \mid |\alpha| = 1\}$.

We define the product of characters $\chi_1 \chi_2$ pointwise. The trivial character is the unit function, the inverse character $\chi^{-1}(a) = \overline{\chi(a)}$.

Definition 1.1. The group of characters of A is called the Pontryagin dual group and is denoted by $\hat{A}$.

Denote the vector space of complex valued functions on A (resp. on $\hat{A}$) by $C(A)$ (resp. by $C(\hat{A})$).

Example 1.2. (1) Any number $c \in \mathbb{C}$ defines a constant function, in particular 1 is the unit function in $C(A)$. Notice that pointwise multiplication makes $C(A)$ into an associative algebra with the unit.
 (2) Any element $a \in A$ gives rise to a delta function δ_a whose value equals 1 at a and 0 at all other elements of A .
 (3) Certainly any character $\chi \in \hat{A}$ provides an element in $C(A)$.

Remark 1.3. It is known from the algebra course that as elements of $C(A)$ all characters are linearly independent. In particular, the cardinality of $\hat{A}$ is no greater than the number of elements in A .

Example 1.4. Take the finite cyclic group $A = \mathbb{Z}/n = \{1, s, \dots, s^{n-1}\}$. Then any $k \in \{0, \dots, n-1\}$ provides a character $\chi_k : \mathbb{Z}/n \rightarrow \mathbb{C}^*, s^m \mapsto \xi^{mk}$. Here ξ denotes a chosen primitive root of unity of degree n . In particular, $\hat{A}$ is a cyclic group of the same order. It is isomorphic to A non-canonically.

Classification Theorem for finite Abelian groups implies that for any such A the group $\hat{A}$ is non-canonically isomorphic to A .

The vector space $C(A)$ (resp. $C(\hat{A})$) is equipped with a Hermitian form:

$$\langle f, g \rangle = \sum_{a \in A} f(a) \overline{g(a)}.$$

Remark 1.5. Notice that the functions δ_a , $a \in A$, form an orthonormal basis of $C(A)$.

Lemma 1.6. Let χ_1 and χ_2 be two different characters of A . Then as elements of $C(A)$ they are orthogonal to each other.

Proof. Let $b \in A$ be an element such that $\chi_1(b) \neq \chi_2(b)$. We have

$$\langle \chi_1, \chi_2 \rangle = \sum_{a \in A} \chi_1(a) \overline{\chi_2(a)} = \sum_{a \in A} \chi_1(ab) \overline{\chi_2(ab)} = \left(\sum_{a \in A} \chi_1(a) \overline{\chi_2(a)} \right) \chi_1(b) \chi_2^{-1}(b).$$

By our choice of $b \in A$, the first factor in the last expression equals to 0. $\square$

Lemma 1.7. The map $A \rightarrow \hat{\hat{A}}$, $a \rightarrow \hat{a}$, $\hat{a}(\chi) = \chi(a)$, provides a canonical isomorphism between A and the double Pontryagin dual group of A .

Proof. We know that the source and the target of the canonical map have the same cardinality. It is clear that the map respects multiplication in the group. Moreover, for any non-unit a , the character $\hat{a}$ is non-trivial. It follows that the map is an injective. Thus it is an isomorphism. $\square$

2. DISCRETE FOURIER TRANSFORM

Definition 2.1. The vector space map

$$F : C(A) \rightarrow C(\hat{A}), \quad F(f)(\chi) = \sum_{a \in A} f(a) \overline{\chi(a)},$$

is called the Fourier transform for the group A .

Example 2.2. Let us calculate some examples:

- (1) $F(\delta_a)(\chi) = \overline{\chi(a)} = \hat{a}^{-1}(\chi)$.
- (2) $F(1)(\chi) = \sum_{a \in A} \overline{\chi(a)}$. It is known that for a non-trivial character such sum equals to 0. The value at χ_0 equals to $\#(A)$. Thus we have $F(1) = \delta_{\chi_0}$.
- (3) More generally, $F(\chi_1)(\chi_2) = \langle \chi_1, \chi_2 \rangle = 0$ for $\chi_1 \neq \chi_2$. We have $F(\chi)(\chi) = \langle \chi, \chi \rangle = \#(A)$. It follows that $F(\chi) = \#(A) \delta_\chi$.

Corollary 2.3. It follows that the Fourier transform $F = F_A$ is an isomorphism of the vector spaces $C(A)$ and $C(\hat{A})$ with its inverse equal to $\#(A) F_{\hat{A}}$.

Remark 2.4. We have checked that an orthogonal basis in $C(A)$ is mapped to an orthogonal basis in $C(\hat{A})$. Thus up to a scalar multiple F is an isometry.

Definition 2.5. Given a map of finite sets $\alpha : X \rightarrow Y$, consider the pull back morphism

$$\alpha^* : C(Y) \rightarrow C(X), \quad \alpha^*(f)(x) = f(\alpha(x))$$

and the push forward morphism

$$\alpha_* : C(X) \rightarrow C(Y), \quad \alpha_*(f)(y) = \sum_{\alpha(x)=y} f(x).$$

Remark 2.6. (1) The pull-back map respects the pointwise product of functions.

(2) For two maps $\alpha : X \rightarrow Y$ and $\beta : Y \rightarrow Z$ we have

$$(\beta \circ \alpha)^* = \alpha^* \circ \beta^*, \quad (\beta \circ \alpha)_* = \beta_* \circ \alpha_*$$

. This is a direct calculation.

Let $\alpha : B \rightarrow A$ be a morphism of finite Abelian groups. Then any character of A defines a character of B . We obtain the Pontryagin dual map $\hat{\alpha} : \hat{A} \rightarrow \hat{B}$.

Lemma 2.7. We have $(\hat{\alpha})^* \circ F = F \circ \alpha_*$.

We introduce the second multiplication on $C(A)$. Denote the multiplication map $A \times A \rightarrow A$ by m .

Definition 2.8. For $f, g \in C(A)$ denote the function on $A \times A$ given by $(a, b) \mapsto f(a)g(b)$ by $f \times g$. The convolution $f \star g = m_*(f \times g)$.

Remark 2.9. (1) The element δ_1 is the unit for the operation.

(2) One checks directly that convolution is associative.

(3) Compare the definition of convolution with the following observation. Denote the diagonal embedding $A \rightarrow A \times A$ by Δ . The pointwise product of functions is given by $f \cdot g = \Delta^*(f \times g)$.

Lemma 2.10. We have $F(f \star g) = F(f) \cdot F(g)$.

Proof. Consider the map $\hat{m} : \hat{A} \rightarrow \hat{A} \times \hat{A}$. We have

$$\hat{m}(\chi)(a, b) = \chi(ab) = \chi(a)\chi(b) = (\chi \times \chi)(a, b).$$

Thus the Pontryagin dual for the product map is the diagonal embedding. The proof follows. $\square$

Consider the argument shift operation $L_a : C(A) \rightarrow C(A)$, $(L_a f)(b) = f(a + b)$.

Lemma 2.11. We have $(F \circ L_a)(f) = \hat{a} \cdot F(f)$.

Proof. The key observation in the proof is that $L_a(f) = \delta_a \star f$. Then use the previous Lemma. $\square$

3. GENERALIZED POISSON SUMMATION

Below we describe the setting for Poisson summation. Let A be a finite Abelian group with a subgroup B . Consider the embedding map $i : B \rightarrow A$ and the projection map $p : A \rightarrow A/B$. The map $\hat{i}$ is surjective, its kernel $B^\perp$ consists of characters χ whose restriction to B is trivial. We identify the kernel with $\widehat{(A/B)}$.

Theorem 3.1. For $f \in C(A)$, for any $a \in A$, we have

$$\sum_{b \in B} f(a + b) = \sum_{\chi \in B^\perp} F(f)(\chi) \bar{\chi}(a).$$

Proof. First notice that LHS in the formula in question equals $p_*(f)(aB)$. From Section 2 we know that the latter equals $F_{A/B}^{-1} \circ (\hat{p})^* \circ F_A$. Notice that

$$F_{A/B}^{-1} \circ (\hat{p})^* \circ F_A(f)(aB) = \sum_{\chi \in B^\perp} F(f)(\chi) \overline{\chi}(a).$$

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